

ERFAN MOHAMMADZADEH

Software Engineer | Data Engineer | C++ / Qt | C# / .NET Web API

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PROFESSIONAL SUMMARY

Software and data engineer specializing in **high-performance C++/Qt applications, sensor and signal pipelines, and structured geospatial reconstruction**. I also build **C# / .NET** backends: **ASP.NET Core Web APIs, REST CRUD endpoints, and database-backed persistence**. I design systems that take raw measurements (cameras, LiDAR, ECG) through filtering, feature extraction, validation, visualization, and export—and expose or store results through structured APIs and SQL. My next step is to deepen this **.NET service and database** work in production web and data platforms.

TECHNICAL SKILLS

Languages: C++, C#, Python, SQL, QML

.NET / backend: ASP.NET Core, Web API, REST CRUD, service-layer APIs

Data & storage: SQL databases, SQLite, XML, structured ETL-style pipelines

Desktop & UI: Qt (Widgets / QML)

Vision & 3D: OpenCV, PCL, VTK, CGAL, Open3D

Geospatial: PDAL, GDAL, QGIS, City4CFD / LoD modeling

Systems: Linux (LPIC-1), cross-platform desktop builds

Domains: ECG / biomedical DSP, camera calibration, ANPR, LiDAR city models

CAREER DIRECTION

Growing toward **C# / .NET backend engineering**: ASP.NET Core Web APIs, CRUD over relational data, and connecting desktop/processing products to maintainable service and database layers.

WORK EXPERIENCE

Geospatial Data Engineer | Image Horizon (Data Horizon) · Tehran, Iran

November 2025 – Present (concurrent with Amvaj Negar)

- Own the **City4CFD / QCity4CFD** reconstruction path: point clouds + building footprints → **LoD 3.0/2.2/2.0/1.3/1.0** city meshes for CFD, covering **20,000+ buildings**.
- Built processing stages with **PCL, PDAL, GDAL, CGAL, and VTK**: filtering, segmentation, surface reconstruction, geometric regularization, rendering, and mesh QA.
- Reported **>90% reconstruction accuracy** on the high-detail building pipeline (Python/NumPy/Open3D/PDAL + GIS in QGIS/ArcGIS).
- Bridged GIS operators and simulation teams by producing inspectable, simulation-ready geometry instead of raw point clouds.

Senior Software Engineer | Amvaj Negar Sepahan Co. · Isfahan, Iran

April 2024 – Present

- Designed and maintain **Holter ECG desktop software** (C++ / Qt Widgets): long-term ECG ingest, **P-Q-R-S-T** detection, beat-template classification, arrhythmia support, **SQLite** persistence, XML/PDF reporting.
- Built **QCardio**, a **validation harness** for ECG libraries: converted **MIT-BIH Arrhythmia** into a structured test set so algorithms are checked against physician-annotated ground truth, not only visual review.
- Own signal-processing correctness: filtering, feature extraction, classification, and regression tests that clinicians and engineers can both trust.

Software Engineer | Data Image Rayan Co. · Isfahan, Iran

June 2024 – March 2025 (concurrent with Amvaj Negar)

- Delivered **ANPR** monitoring for parking access: live cameras, vehicle detection, plate crop, OCR validation, and barrier control.
- Reduced manual gate intervention by closing the loop from video frame to access decision with a real-time image pipeline.

Junior Software Engineer | Tivan Sanat (Dade Pardazan Tivan Sanat) · Isfahan, Iran

April 2021 – December 2022

- Shipped a cross-platform **C++/Qt + OpenCV** camera-calibration tool: chessboard/target detection, keypoints, **intrinsics/extrinsics**, distortion (radial/tangential), focal length, principal point, FOV, and correction matrices.
- Exported calibration results (SQLite / XML / PDF) for production optical QA.

SELECTED PROJECTS

ASP.NET Core Web API (CRUD) — current backend practice

REST endpoints for create / read / update / delete against a SQL database; request handling, data access, and API structure for service-oriented products.

C#, ASP.NET Core, Web API, SQL

QCity4CFD — Data Horizon · Jan 2025

LoD 2.2 reconstruction from LiDAR + footprints; mesh regularization and CFD-oriented city models.

Python, Open3D, PDAL, CGAL, QGIS/ArcGIS, City4CFD

QCardio — Amvaj Negar Sepahan · Jun 2026

ECG library validation against annotated MIT-BIH records; automated pass/fail on clinical waveforms.

ECG Holter Software — Amvaj Negar Sepahan · Apr 2024

Desktop Holter analysis: wave detection, templates, arrhythmia flags, reports.

Visible-camera parameter tester — Tivan Sanat · Dec 2022

Production calibration of practical camera parameters for accurate rendering and metrology.

EDUCATION

B.Sc. Electrical Engineering — Communication Systems

Semnan University · Semnan, Iran · Oct 2019 – Jan 2024 · GPA 3.2 / 4.0

Coursework and practice in signal processing, communications, and control; applied DSP (filtering, features, real-time classification) in subsequent industry systems.

CERTIFICATES

Foundations of Coding: Full-Stack — Microsoft / Coursera · Oct 2025

<https://www.coursera.org/account/accomplishments/verify/XIAL95E2ZNBP>

LANGUAGES

Persian (Farsi): Native

English: Professional working proficiency (reading, writing, speaking, listening) — used for technical documentation, code, and client-style communication.